

PIMPRI CHINCHWAD EDUCATION TRUST'S  
**PIMPRI CHINCHWAD COLLEGE OF ENGINEERING**  
SECTOR NO. 26, PRADHIKARAN, NIGDI, PUNE – 411044



## **B. TECH PROJECT SYNOPSIS**

on

**Title: “Cafeteria Management System”**

**Project Group Members:**

| <b>Sr. No.</b> | <b>Name of the Student</b> | <b>PRN No.</b> |
|----------------|----------------------------|----------------|
| 1              | Yash Unhale                | 123B1E110      |
| 2              | Varad Wakde                | 123B1E115      |
| 3              | Nutan Shinde               | 123B5E188      |

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Project Guide

**DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING**  
**B.TECH.**  
(E&TC) 2026-27

## Abstract

Traditional cafeteria operations often depend on manual ordering, billing, and record-keeping processes, which can result in long queues, order errors, inefficient management, and delays in service, particularly during peak hours. Customers generally need to wait at the cafeteria counter to view available food items and place their orders, while cafeteria staff have to manage orders and customer information manually. These limitations create a need for a centralized and efficient digital solution for cafeteria management.

The **Web-Based Cafeteria Management System** is proposed as an online platform that digitizes the major activities involved in cafeteria operations. The system allows customers to register and log in, browse available food items and categories, select required items, view the total amount, and make payment before their order is confirmed. After successful payment verification, the order is confirmed and the relevant order and billing details are stored in the database. On the administrative side, the system provides facilities to manage food items, categories, customer information, orders, and payment records through a centralized interface.

The system is developed using **HTML, CSS, and JavaScript** for the frontend, **PHP** for backend processing, and **MySQL** for database management. **Apache/XAMPP** can be used as the local web-server environment. The proposed system provides a simple and user-friendly interface while reducing dependency on manual order processing and improving the transparency of cafeteria transactions.

The expected outcome of the project is a functional web-based cafeteria management platform that improves ordering efficiency, reduces human errors, minimizes waiting time, and provides a secure **Pay First** ordering process. The system can be applied in **college cafeterias, office cafeterias, food courts, restaurants, and institutional canteens**. Future enhancements can include QR-based ordering, mobile application support, inventory management, real-time notifications, food recommendations, and advanced analytics.

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## 1. Group Details and Title of the Project

### Project Title: “Cafeteria Management System”

Web-Based Cafeteria Management System: A Centralized Platform for Online Food Ordering, Menu Management, Order Processing, and Customer Feedback.

### Group Members

| Sr. No. | Name of Student | PRN No.   | Branch | Mobile No. | College Email ID                                                               |
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### Guide and Group Information

|                      |                                             |
|----------------------|---------------------------------------------|
| Guide Name           | Mr. Atul Pawar                              |
| Department           | Electronics & Telecommunication Engineering |
| Group ID / Team Name | GC14                                        |

## 2. Problem Statement

Traditional cafeteria operations often rely on manual processes for user registration, food selection, order placement, billing, and record management. Customers may have to depend on direct interaction with cafeteria staff to select food items and place orders, which can become inconvenient during busy hours and may lead to delays or errors in order processing.

Managing customer details, available food items, orders, and bills manually also makes it difficult for cafeteria administrators to maintain accurate and organized records. In addition, the absence of a centralized system for collecting customer feedback limits the ability of cafeteria management to understand customer experiences and improve services.

The **Cafeteria Management System** addresses these issues by providing a web-based platform that integrates customer registration and login, cafeteria user management, menu/item display, online order placement, bill generation, and feedback collection into a single system. The administrator can manage users, food items, and order records through a centralized interface, while customers can access the available menu and place their orders through the website.

Thus, the proposed system aims to **reduce manual effort, organize cafeteria records, simplify the ordering process, minimize errors, and improve the overall efficiency of cafeteria management** through a centralized web-based solution.

### **3. Objective and Scope of the Project**

#### **Objectives**

The main objectives of the Cafeteria Management System are:

1. To develop a simple web-based system for managing cafeteria food items and orders.
2. To allow customers to browse the menu, place orders, and make payment before order confirmation.
3. To help administrators easily manage food items, customers, orders, and payments.
4. To reduce manual work, waiting time, and errors in cafeteria operations.

#### **Scope and End Users**

The scope of the Cafeteria Management System covers the digital management of major cafeteria activities, including user registration and authentication, menu browsing, food-item management, online order placement, order management, and feedback collection. The system provides a centralized platform where customers can interact with cafeteria services and administrators can manage the operational records.

The system can be used in college cafeterias, university canteens, office cafeterias, institutional food courts, and similar food-service environments.

Primary End Users:

- Customers/Students: Register, log in, browse food items, place orders, and provide feedback.
- Cafeteria Administrator: Manage food items, categories, users, and orders.
- Cafeteria Staff: Process and manage customer orders according to the system records.

The system focuses on improving the speed, accuracy, organization, and convenience of cafeteria operations through web-based automation.

## 4. Methodology

The **Cafeteria Management System** follows a structured web-based workflow that integrates customer interaction, menu management, online payment, order processing, billing, and feedback management into a centralized platform.

The user first accesses the system and registers or logs in to the platform. After successful authentication, the customer can browse the available food items and categories displayed by the cafeteria. The customer selects the required food items and quantity, after which the system calculates and displays the total amount. The customer then proceeds with the **Pay First** option. The payment is verified before the order is confirmed. Only after successful payment is the order stored in the database and confirmed for processing.

On the administrative side, the administrator can manage food items, categories, user information, orders, and payment records through the management interface. The administrator can view and process confirmed customer orders and maintain updated cafeteria records. The system also provides a feedback facility through which customers can submit their suggestions or experiences.

The overall methodology can be summarized as:

**User Registration/Login → Browse Menu → Select Food Items → Calculate Total → Pay First → Payment Verification → Order Confirmation → Store Order Details in MySQL → Generate Bill → Admin Processes Order → Customer Provides Feedback**

The system is implemented using **HTML, CSS, and JavaScript** for the user interface, **PHP** for server-side processing and business logic, and **MySQL** for storing and managing application data. This integrated approach reduces manual operations, improves payment and order management, maintains organized records, and provides a convenient platform for both customers and cafeteria administrators.

## 5. Hardware and Software Requirements

The **Cafeteria Management System** is a web-based application and does not require any specialized hardware. The system can be executed on a standard computer with a local web server and MySQL database environment. The technologies identified from the project source code include **PHP, HTML, CSS, JavaScript, MySQL, and Apache/XAMPP-compatible local hosting**. The project uses PHP's mysqli extension to connect to a MySQL database named cafeteria.

| Category                | Technologies Used                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend                | HTML, CSS, JavaScript                                                                                                                           |
| Backend                 | PHP                                                                                                                                             |
| Database                | MySQL                                                                                                                                           |
| Database Connectivity   | PHP mysqli                                                                                                                                      |
| Web Server              | Apache                                                                                                                                          |
| Development Environment | XAMPP / Local PHP Server                                                                                                                        |
| Hardware Requirements   | Standard computing device (laptop/desktop) with internet connectivity; no special hardware is required as the system is entirely software-based |

## 6. Application and Future Scope of the Project

### Applications

The **Cafeteria Management System** can be applied in various food-service environments where efficient management of menus, customers, orders, and payments is required.

- **College and University Cafeterias:** Students and staff can browse available food items, make payment, and place orders conveniently.
- **Office Cafeterias:** Employees can view the menu and make prepaid food orders without depending entirely on manual ordering.
- **Institutional Canteens:** The system can help manage customer orders, payments, and cafeteria records efficiently.
- **Food Courts:** Food items and categories can be managed through a centralized web-based platform.
- **Small Restaurants and Cafeterias:** The system can reduce manual order management and maintain customer and transaction information digitally.

### Future Scope

The proposed system can be further enhanced with the following features:

- **QR Code-Based Ordering:** Customers can scan a QR code at the cafeteria counter or table to directly access the menu and place orders.
- **Mobile Application:** Development of Android/iOS applications for convenient access to cafeteria services.
- **Inventory Management:** Automatic tracking of food ingredients and stock levels to help administrators manage inventory efficiently.
- **Real-Time Order Notifications:** Notifications can be provided to customers when their order is accepted, prepared, or ready for collection.
- **Order History and Analytics:** Generation of reports and analytics based on previous orders, popular food items, and customer activity.
- **Food Recommendation System:** Personalized food recommendations can be introduced based on customer preferences and previous orders.
- **Advanced Payment Integration:** Support for multiple payment methods such as UPI, cards, and digital wallets with improved transaction verification.
- **Digital Billing and Receipts:** Automated generation of digital bills and receipts for successfully completed transactions.

## 7. Project Timeline

The project has been planned and executed over a period of approximately 14 weeks, divided into the following phases:

| Weeks      | Phase                                      | Deliverable                                                                            |
|------------|--------------------------------------------|----------------------------------------------------------------------------------------|
| Week 1-2   | Requirement Analysis & Problem Definition  | Finalized problem statement, objectives, scope, and system requirements                |
| Week 3-4   | System Design & Technology Stack Selection | System architecture, database design, and user-interface design                        |
| Week 5-6   | Database & Backend Development             | MySQL database, PHP backend, user authentication, and database connectivity            |
| Week 7-8   | Frontend Development                       | Customer interface, menu display, food categories, and navigation                      |
| Week 9-10  | Order & Admin Module Development           | Cart/order functionality, order management, food-item management, and admin operations |
| Week 11-12 | Integration & Testing                      | Frontend-backend integration, database testing, functional testing, and bug fixing     |
| Week 13    | Documentation                              | Project report, synopsis, diagrams, screenshots, and technical documentation           |
| Week 14    | Final Demonstration & Submission           | Final testing, project demonstration, presentation, and submission                     |

## 8. References

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